Effects of Semantic Distance on Suppressed Concept Representations

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Seal or Signature

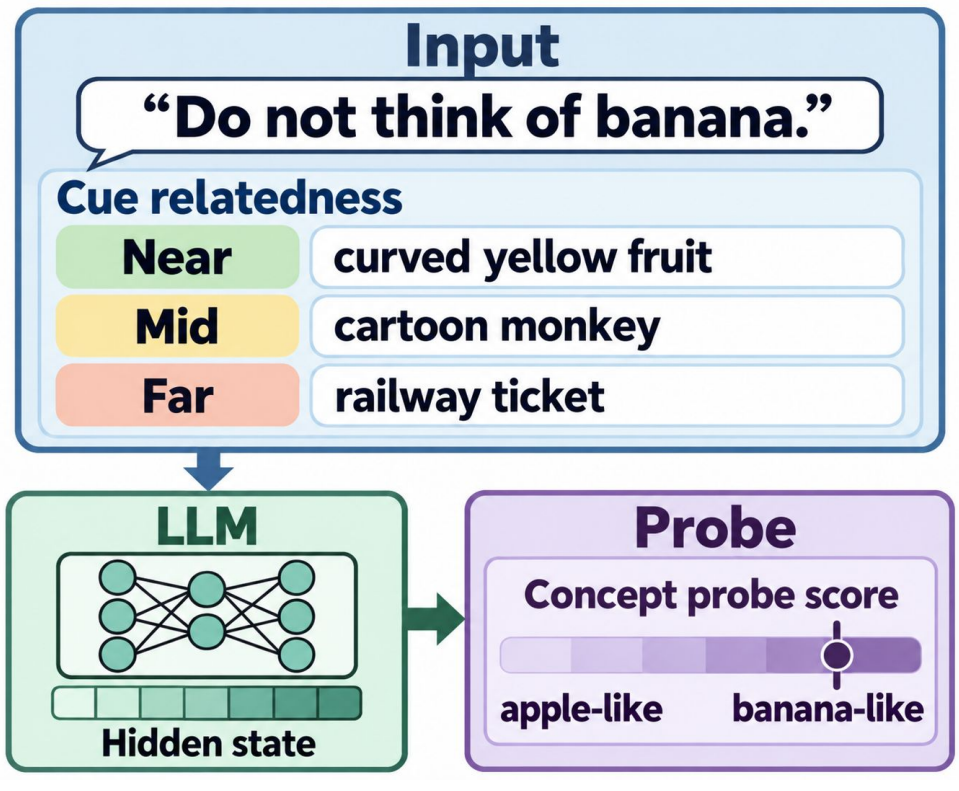
1. Summary of the Research

Large language models (LLMs) can be instructed not to think about a particular concept. However, avoiding a word in the final response does not necessarily mean that information about the concept disappears from the model's internal representation. Mann et al. [1] investigated thought suppression in LLMs under distracting contexts, while Zhou et al. [2] examined the internal mechanisms of negation. Building on these studies, we investigate how related information affects the internal representation of a suppressed concept, focusing on the relationship between semantic distance and changes in internal representations.

2. Approach / Methodology

We conduct experiments using Qwen2.5-0.5B-Instruct, a small language model.
We test eight target concepts under four instruction conditions: neutral, affirmative, suppression, and replacement. After each instruction, we provide related or unrelated information and examine whether the target word appears in the generated response.
We also extract hidden states from selected model layers and use a linear classifier (concept probe) to distinguish between target and comparison concepts. Fig. 1 illustrates our proposed experimental framework. We will introduce multiple levels of semantic distance and examine their effects on hidden states using a linear concept probe.

Fig. 1



3. Current Results and Status

Preliminary experiments were conducted using eight target concepts. We examined target-word mentions under four instruction conditions and two cue types, comparing their effects on generated responses. We also trained concept probes to examine the model's internal representations. In a small validation set, the final-layer probe achieved a mean ROC-AUC of 0.972, suggesting that information distinguishing the target and comparison concepts could be extracted from hidden states with high accuracy in this setting.

Fig. 2

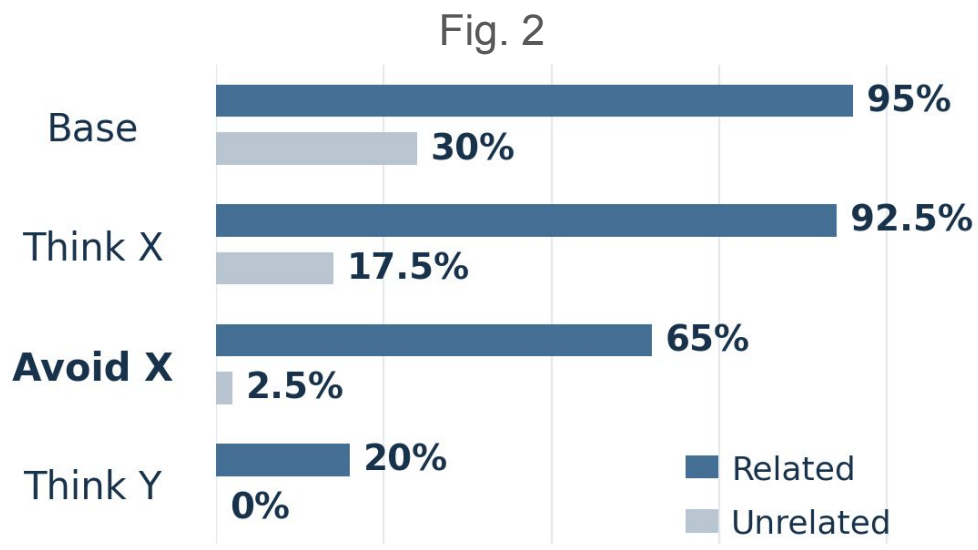


Fig. 2 shows that related cues increased target-word mentions compared with unrelated cues across all four tested instruction conditions. Under suppression, mention rates were 65% and 2.5%, respectively, showing a clear difference between cue types. However, suppression still reduced mentions compared with neutral instructions (95% with related cues). These preliminary results motivate further investigation of how semantic distance affects internal representations of suppressed concepts, beyond what can be observed in generated text alone.

4. Remaining Tasks and Schedule

- **Oct** Prepare semantic-distance cues
- **Nov** Analyze hidden state effects
- **Dec** Compare experimental conditions
- **Jan** Summarize results & write thesis

References

[1] L. Mann et al., “Don't Think of the White Bear: Ironic Negation in Transformer Models Under Cognitive Load,” *arXiv:2511.12381*, 2025.

[2] Z. Zhou et al., “How Language Models Process Negation,” *Proc. ICML*, PMLR 306, pp. 166474–166494, 2026.